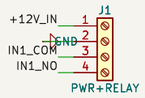
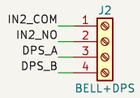
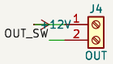
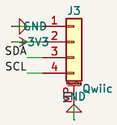
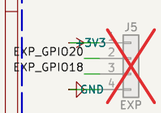
CONNECTORS (field side)



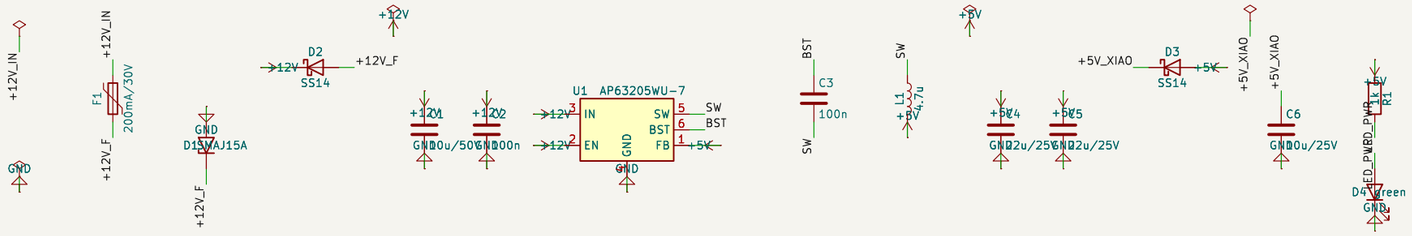








POWER: 12 V input protection, 5 V buck, XIAO 5 V feed



Power budget: UA-Hub LINE output 12 V / 0.33 A max. F1 holds 200 mA.

D2 = reverse-polarity protection; D1 + F1 clamp surges / reversed input.

D3: Seeed requires a diode (anode = supply) when feeding the XIAO 5V pin.

Field loops: +12V -> 2.2k -> opto LED -> COM ... contact ... NO -> 1k -> GND (~3.3 mA).

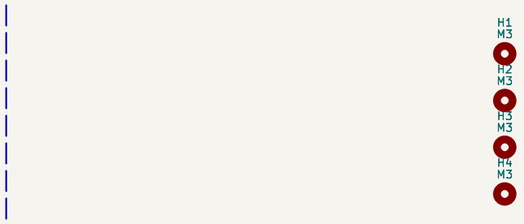
Contact closed -> opto ON -> INx_SIG LOW (active low). RC 1k/10n + firmware debounce.

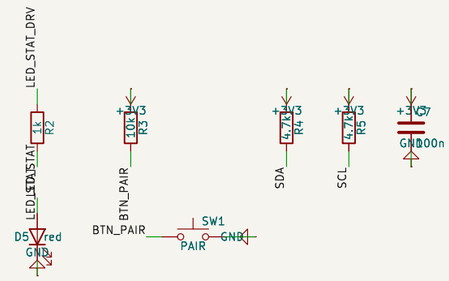
J2.3/J2.4 (DPS): JP1/JP2 1-2 = input (reed contact); 2-3 = SSR output to UA-Hub DPS input.

Never wire DPS input in parallel with a reed that already feeds the UA-Hub DPS input.

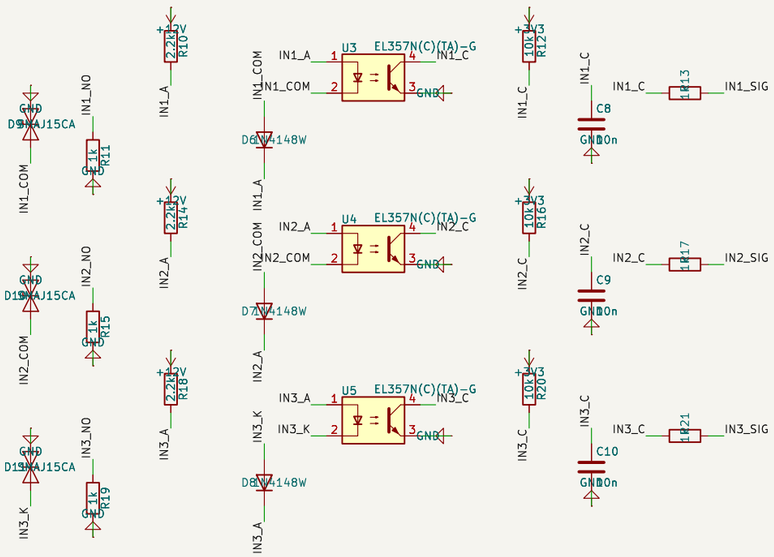
No mains voltage on this PCB. 230 V doorbells: use a certified coupling relay (electrician).

MCU: XIAO ESP32C6 (socketed)

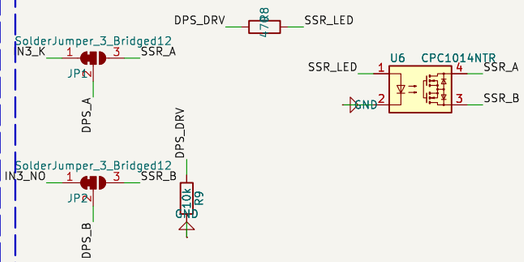




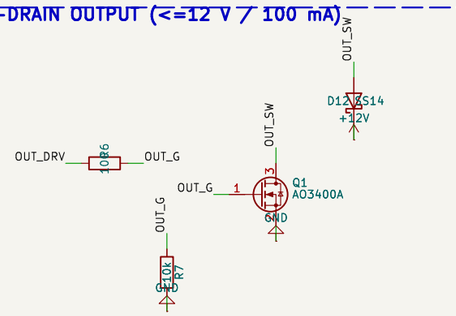
CONTACT INPUTS: opto loops powered from board +12 V (no external voltage)



DPS PORT: Input (default) or isolated SSR output



OPEN-DRAIN OUTPUT (<=12 V / 100 mA)



License: CERN-OHL-S-2.0

Zigbee dry-contact bridge for UniFi Access UA-Hub, Seeed XIAO ESP32C6

AccessBridge project

Sheet: /

File: accessbridge.kicad_sch

Title: AccessBridge C6

Size: A3	Date: 2026-10-18	**Rev: A (HW-1.0)**
KiCad E.D.A. 9.0.9		Id: 1/1